

EXPERIMENT 26. Machine Translation Using Hugging Face Transformers

AIM:

To implement a machine translation program using the Hugging Face Transformers library to translate English text into French using Python.

PROGRAM:

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

model_name = "Helsinki-NLP/opus-mt-en-fr"

tokenizer = AutoTokenizer.from_pretrained(model_name)

model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

text = "Artificial Intelligence is changing the world."

inputs = tokenizer(text, return_tensors="pt")

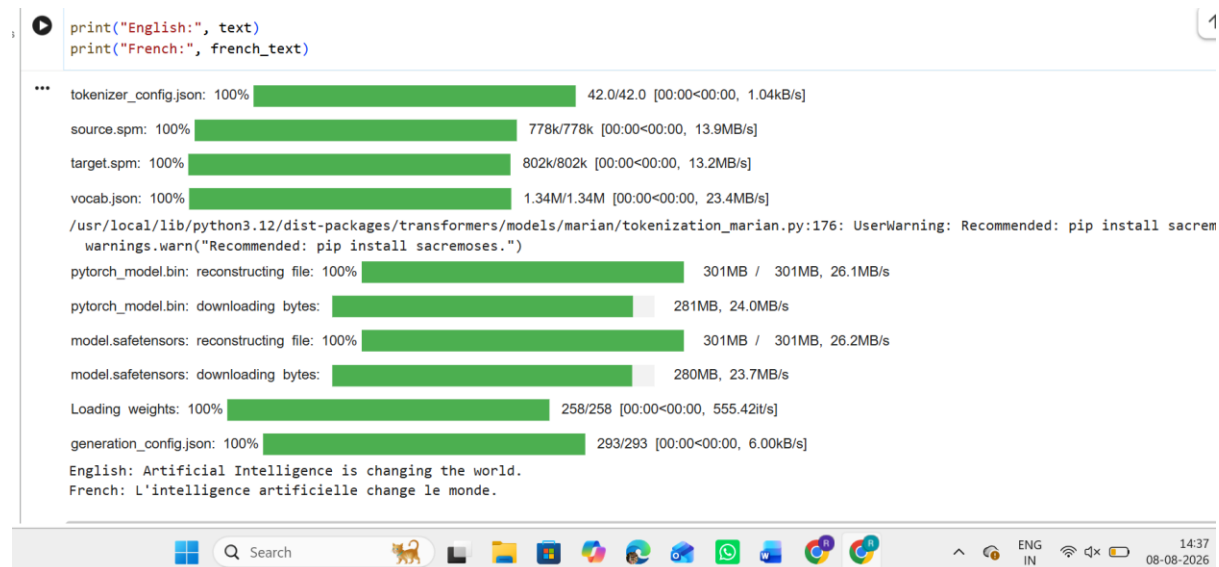
translated = model.generate(**inputs)

french_text = tokenizer.decode(translated[0], skip_special_tokens=True)

print("English:", text)

print("French:", french_text)
```

OUTPUT:



```
print("English:", text)
print("French:", french_text)
```

tokenizer_config.json: 100% 42.0/42.0 [00:00<00:00, 1.04kB/s]

source.spm: 100% 778k/778k [00:00<00:00, 13.9MB/s]

target.spm: 100% 802k/802k [00:00<00:00, 13.2MB/s]

vocab.json: 100% 1.34M/1.34M [00:00<00:00, 23.4MB/s]

/usr/local/lib/python3.12/dist-packages/transformers/models/ Marian/tokenization_marian.py:176: UserWarning: Recommended: pip install sacremoses.warn("Recommended: pip install sacremoses.")

pytorch_model.bin: reconstructing file: 100% 301MB / 301MB, 26.1MB/s

pytorch_model.bin: downloading bytes: 281MB, 24.0MB/s

model.safetensors: reconstructing file: 100% 301MB / 301MB, 26.2MB/s

model.safetensors: downloading bytes: 280MB, 23.7MB/s

Loading weights: 100% 258/258 [00:00<00:00, 555.42it/s]

generation_config.json: 100% 293/293 [00:00<00:00, 6.00kB/s]

English: Artificial Intelligence is changing the world.

French: L'intelligence artificielle change le monde.